Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2024-25 (ODD)	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	50	Name of Student	Devashish Dhote

Practical Number	**6**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**JavaScript — Arrays, Objects & Higher-Order Functions.**
Problem Definition	**Develop a JavaScript program to model a shopping cart using an array of product objects. Each product should contain id, name, price, and quantity.**
Theory (100 words)	**This shopping cart model demonstrates the core principles of state management and dynamic rendering in modern frontend development.** The JavaScript array of objects acts as the **single source of truth** (the state), encapsulating all necessary product data. Instead of hardcoding visual elements, we use declarative **higher-order functions** like map() and reduce(). These functions process the data immutably, generating dynamic values and HTML strings on the fly. Finally, JavaScript bridges the gap between logic and UI by manipulating the **Document Object Model (DOM)**. This ensures that the HTML interface always accurately reflects the underlying data structure in real time.
Procedure and Execution (100 Words)	**Step for Implementation:** **Define the Data State:** Create a JavaScript array of objects to store the shopping cart's underlying data (ID, name, price, and quantity for each item). • **Build the HTML Skeleton:** Write the base HTML document with empty container `<div>` elements (using unique `id` attributes) where the dynamic content will eventually go.

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	• **Style the Interface:** Apply CSS rules to format the layout, using Flexbox to align item names on the left and prices on the right for a clean receipt look. • **Target the DOM:** In the JavaScript file, use `document.getElementById()` to select the empty HTML containers so the script knows exactly where to inject the data. • **Generate UI Elements:** Use the `map()` higher-order function to loop through the data array, transforming each product object into a customized block of HTML code. • **Calculate the Total:** Use the `reduce()` function to iterate through the cart array, multiplying each item's price by its quantity and adding it to a running mathematical total. • **Render to the Webpage:** Assign the generated HTML strings and the calculated total to the `innerHTML` properties of your targeted DOM elements, forcing the browser to instantly display the fully populated cart.
CODE:	**HTML:** <!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>JS Shopping Cart</title> <link rel="stylesheet" href="styles.css"> </head> <body> Your Shopping Cart <!-- JS will insert the cart items here --> <!-- JS will insert the total price here --> <script src="script.js"></script> </body>

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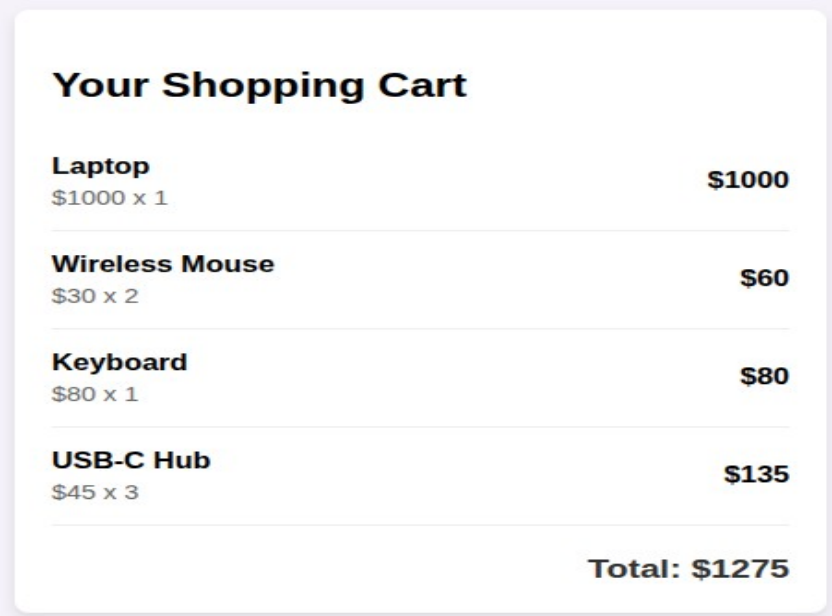
	</html> **JS:** const cart = [{ id: 1, name: "Laptop", price: 1000, quantity: 1 }, { id: 2, name: "Wireless Mouse", price: 30, quantity: 2 }, { id: 3, name: "Mechanical Keyboard", price: 80, quantity: 1 }, { id: 4, name: "USB-C Hub", price: 45, quantity: 3 }]; const cartItemsContainer = document.getElementById("cart-items"); const cartTotalContainer = document.getElementById("cart-total"); function displayCart() { const htmlString = cart.map(product => { const itemTotal = product.price * product.quantity; return ` <div class="cart-item"> <div> <h4>${product.name}</h4> <p>$${product.price} x ${product.quantity}</p> </div> <strong>$${itemTotal}</strong> </div> `; }).join(''); cartItemsContainer.innerHTML = htmlString; } function displayTotal() { const finalTotal = cart.reduce((total, product) => { return total + (product.price * product.quantity); }, 0); cartTotalContainer.innerHTML = `<strong>Total: $${finalTotal}</strong>`; } displayCart(); displayTotal();
Output Analysis	**Visual Output:** The browser displays a styled, receipt-like list showing each product, its individual subtotal, and a final combined total at the bottom.

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	Data Processing: • **Item Subtotals (map()):** The program calculates the cost for each row (e.g., 2 Wireless Mice × $30 = **$60**) and generates the HTML to display it. • **Final Total (reduce()):** The program adds all individual subtotals together (1000 + 60 + 80 + 135) to output the final cart value of **$1275**. 
Link of student Github profile where lab assignment has been uploaded	**https://github.com/devdhote20**
Conclusion	The output is a direct reflection of the underlying data state. Because the calculations arehandled dynamically by JavaScript rather than hardcoded in HTML, changing any quantity or price in the initial `cart` array will instantly update both the individual item lines and the final `$1275` total on the screen.

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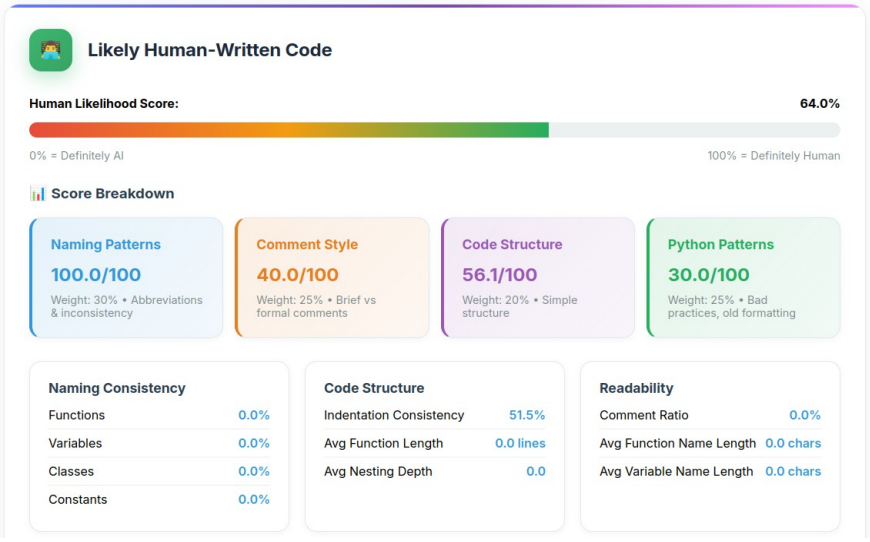
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